B.M.S. COLLEGE OF ENGINEERING

(AUTONOMOUS COLLEGE UNDER VTU)

BULL TEMPLE ROAD, BASAVANGUDI, BANGALORE - 560019

a Project Management Capstone Report on

VISIONQAI: AI-BASED VISUAL QUALITY INSPECTION FOR BOTTLE CAPS, FILL LEVEL AND PRODUCT LABELS

(Entrepreneurship and Project Management)

in fulfillment of the requirements for the award of degree

BACHELOR OF ENGINEERING

in

Computer Science & Engineering (Data Science)

by

Sushanth S (1BM23CD063)

Ullas N (1BM23CD067)

Shreyas Bairy K S (1BM23CD058)

Under the Guidance of

Dr. Shambhavi B R

Professor

Department of CSE (Data Science),

B.M.S. College of Engineering

AY 2026-2027

B.M.S. COLLEGE OF ENGINEERING

(Autonomous College under VTU)

Bull Temple Road, Basavangudi, Bangalore - 560019

Department of Computer Science and Engineering (Data Science)

CERTIFICATE

This is to certify that the project entitled "**VisionQAI: AI-Based Visual Quality Inspection for Bottle Caps, Fill Level and Product Labels**" is a bona fide work carried out by the following students for the course Entrepreneurship and Project Management. It is certified that all corrections and suggestions indicated during the internal assessments have been incorporated in the final report submitted to the department. The project report has been approved as it meets the academic requirements prescribed for the Bachelor of Engineering Degree in respect of this course.

Candidate details

SL. NO.	Student Name	USN
1	Sushanth S	1BM23CD063
2	Ullas N	1BM23CD067
3	Shreyas Bairy K S	1BM23CD058

Dr. Shambhavi B R	**Dr. Shyamala G**	**Dr. Bheemsha Arya**
Professor	HOD	Principal

Examiners

Name of Examiner	**Signature of the examiner**
1.	_____
2.	_____

B.M.S. COLLEGE OF ENGINEERING

(AUTONOMOUS COLLEGE UNDER VTU)

BULL TEMPLE ROAD, BASAVANGUDI, BANGALORE - 560019

DECLARATION

We hereby declare that the project report entitled ***"VisionQAI: AI-Based Visual Quality Inspection for Bottle Caps, Fill Level and Product Labels"*** is a bona fide work carried out by us under the guidance of **Dr. Shambhavi B R**, Professor, with **Prof. Nayana N Kumar**, Assistant Professor, as our project guide, Department of Computer Science and Engineering (Data Science), B.M.S. College of Engineering, Bengaluru, in partial fulfillment of the requirements of the degree of Bachelor of Engineering in Computer Science & Engineering (Data Science) of Visvesvaraya Technological University, Belagavi. We further declare that, to the best of our knowledge and belief, this report has not been submitted either in part or in full to any other university or college.

Candidate details

SL. NO.	Student Name	USN	Student's Signature
1	Sushanth S	1BM23CD063	
2	Ullas N	1BM23CD067	
3	Shreyas Bairy K S	1BM23CD058	

Date: 02-11-2026

Acknowledgments

The satisfaction that accompanies the successful completion of this ***Capstone Project*** would be incomplete without the mention of the people who made it possible through constant guidance and encouragement.

We express our heartfelt gratitude to **Dr. B. S. Ragini Narayan**, *Chairperson*, Donor Trustee, Member Secretary & Chairperson, BMSET; **Dr. P. Dayananda Pai**, *Member Life Trustee*, BMSET; and **Dr. Bheemsha Arya**, *Principal*, B.M.S. College of Engineering, for providing the infrastructure required to complete this project.

We are grateful to **Dr. Shyamala G**, Head of the Department, Computer Science and Engineering (Data Science), and to the project coordinators for their constant support.

We express our sincere thanks to **Dr. Shambhavi B R**, Professor, Department of Computer Science and Engineering (Data Science), under whose guidance this report was prepared, and to our project guide, **Prof. Nayana N Kumar**, Assistant Professor, Department of Computer Science and Engineering (Data Science), for helping us throughout and guiding us from time to time.

We also thank all faculty members of the Department of Computer Science and Engineering (Data Science) for their views and encouraging ideas.

Sushanth S (1BM23CD063)
Ullas N (1BM23CD067)
Shreyas Bairy K S (1BM23CD058)

Contents

Certificate i

Declaration ii

Acknowledgments iii

List of Tables vi

List of Figures vii

Project Details and Basis of the Figures viii

1 Executive Summary 1

2 Project Foundation 2

2.1 Project Attributes . . . 2
2.2 Balancing Project Constraints . . . 2
2.3 Project Identification and Selection . . . 3
2.4 Stakeholder Engagement . . . 3
2.5 Project Charter . . . 4
2.6 Request for Proposal and Soliciting Proposals . . . 4
2.7 Life Cycle and Planning Process . . . 4

3 Project Planning 6

3.1 Project Objective . . . 6
3.2 Project Scope Document . . . 6
3.3 Plan for Quality . . . 6
3.4 Work Breakdown Structure . . . 7
3.5 Responsibility Assignment Matrix . . . 8

4 Developing the Schedule 8

4.1 Activity List, Resources and Durations . . . 8
4.2 Network Diagram and Schedule Calculations . . . 9
4.3 Milestones . . . 10
4.4 Baseline Schedule . . . 11
4.5 Project Control Process . . . 11
4.6 Effects of Actual Schedule Performance . . . 12
4.7 Incorporating Changes and Updating the Schedule . . . 13

5 Determining Cost and Budget 13

5.1 Estimate Activity Costs and the Total Budgeted Cost . . . 13
5.2 Time-Phased Budget (Cumulative Budgeted Cost) . . . 14

5.3 Actual Cost and Committed Cost . . . 15
5.4 Earned Value of the Work Performed . . . 16
5.5 Analyse Cost Performance . . . 17
5.6 Forecast the Cost at Completion . . . 18

6 Managing Risk **20**
6.1 Identify and Assess Risks . . . 20
6.2 Plan Risk Responses . . . 20
6.3 Monitor Risks . . . 21

7 The Project Team **22**
7.1 Team Effectiveness . . . 22
7.2 Conflict and Problem Solving . . . 22

8 Lessons Learned and Conclusion **23**

9 Individual Contribution and Reflection **24**

Consistency Checks **26**

Appendix A: WBS Diagram **27**

Appendix B: Network Diagram and Detailed Gantt Chart **27**

Appendix C: Status Summaries **28**

Appendix D: Cost Workings, Quotations and Receipts **29**

Appendix E: Test Results and Other Project Artifacts **29**

List of Tables

Table 0.3 - Rates used to value effort and purchases . . . viii
Table 2.1 - Project attributes evidenced . . . 2
Table 2.2 - Constraint prioritisation . . . 2
Table 2.3 - Project selection scoring . . . 3
Table 2.4 - Assumptions behind each option . . . 3
Table 2.5 - Stakeholder register . . . 3
Table 2.6 - Project charter . . . 4
Table 2.7 - The project on its life cycle . . . 4
Table 3.1 - Project scope document . . . 6
Table 3.2 - Quality plan . . . 6
Table 3.3 - Work breakdown structure with work package budgets . . . 7
Table 3.4 - Responsibility assignment matrix . . . 8
Table 4.1 - Activity list with resource and duration estimates . . . 8
Table 4.2 - Forward and backward pass, slack and critical path . . . 9
Table 4.3 - Milestones, all in 2026 . . . 10
Table 4.4 - Schedule variance analysis . . . 12
Table 4.5 - Change request log . . . 13
Table 5.1 - Activity cost estimates and total budgeted cost . . . 13
Table 5.2 - Time-phased budget . . . 14
Table 5.3 - Actual and committed cost as of the Status Date . . . 15
Table 5.4 - Cumulative earned value . . . 16
Table 5.5 - Cost performance as of the Status Date . . . 17
Table 5.6 - Forecasted cost at completion . . . 18
Table 5.7 - Interim cost forecast, 31 July 2026 . . . 18
Table 5.8 - Monthly control data behind Figure 5.1 . . . 19
Table 6.1 - Risk identification and assessment . . . 20
Table 6.2 - Risk assessment matrix: response plans . . . 20
Table 6.3 - Risk monitoring record . . . 21
Table 7.1 - Team effectiveness . . . 22
Table 7.2 - Problem-solving record . . . 22
Table 8.1 - Reflection on the tools applied . . . 23
Table 9.1 - Individual contribution . . . 24
Table 10.1 - Consistency checks . . . 26
Table C.1 - Monthly status from the commit record . . . 28
Table E.1 - Recorded results . . . 29

List of Figures

Figure 4.1 - Baseline schedule 11
Figure 5.1 - Cumulative budgeted, actual and earned value curves 17
Figure A.1 - Work breakdown structure 27
Figure B.1 - Activity-on-node network 27
Figure B.2 - Baseline against actual for each activity 28

PROJECT MANAGEMENT

Capstone Project Report

Application of Project Management Concepts

Field	Details
Project Title	VisionQAI: AI-based visual quality inspection for bottle caps, fill level and product labels (earlier name: VisionFood QAI)
Team ID	25
Team Members (Name, Roll No., Role)	Sushanth S, 1BM23CD063, Project Manager Ullas N, 1BM23CD067, Models Lead Shreyas Bairy K S, 1BM23CD058, Platform and Dashboard Lead
Course	Entrepreneurship and Project Management
Report Guide	Dr. Shambhavi B R, Professor, Department of CSE (Data Science), B.M.S. College of Engineering
Faculty Mentor (Project Guide)	Prof. Nayana N Kumar, Assistant Professor, Department of CSE (Data Science), B.M.S. College of Engineering
Head of the Department	Dr. Shyamala G, Department of CSE (Data Science), B.M.S. College of Engineering
Submission Date	2 November 2026
Report Status Date	26 October 2026
Project Start - Planned Finish	2 February 2026 - 30 September 2026 (actual finish 26 October 2026)

Project details

Basis of the figures in this report

For this report, the baseline schedule, the budget and all hours were reconstructed retrospectively from the Git history (157 commits across six branches), the project's planning documents and the team's recollection. Actual finish dates cite commits wherever one exists. All cost is notional, valued at the course rate card: no cash was spent. Where a figure is an estimate, its table says so.

Resource	Rate	Applied to
Student team member	₹200 per hour	Ullas N, Shreyas Bairy K S
Student team lead	₹250 per hour	Sushanth S
Faculty mentor	₹1,000 per hour	Nayana N Kumar, hours used only
Cloud, hardware, licences, printing, travel	Actual amount	₹0; GPU and facility provided by the college

Table 0.3 - Rates used to value effort and purchases

1. Executive Summary

Manual inspection cannot keep pace with fast food and beverage packaging lines. VisionQAI, built for our department as academic customer and for plant quality-assurance operators as end users, inspects products with computer vision and gives each one a traceable verdict. It has three inspection lines: bottle caps (YOLOv11m-seg student, 6 classes, 3 cameras), fill level (two YOLO detectors) and tube label anomalies (template matching with EfficientAD, on its own dashboard).

The objective was to deliver these lines, with an operator interface and audit trail, between 2 February and 30 September 2026, for a total budgeted cost (TBC) of ₹1,74,500 notional, with cash spending of ₹0.

As of the Status Date, 26 October 2026, the project is 100% complete and finished 18 working days late. The cost performance index (CPI) is 0.921 and the cost variance is -₹15,000: 710 student hours were used against 649 planned. The overrun of ₹15,000 was covered by the ₹17,450 contingency reserve, leaving ₹2,450.

The top two risks by likelihood and impact were repeated model iteration on scarce data (R1), which occurred on both the cap and label lines, and GPU-specific TensorRT deployment (R2), which caused the largest single slip, 26 days on Docker and TensorRT deployment.

The most important lesson is that scope should be cut early and on evidence. Our April plan promised four package types, QR validation and severity grading in 14 days. Delivery became realistic only after we narrowed to what we could test, and a conference rejection for overclaiming confirmed that choice.

2. Project Foundation

2.1 Project Attributes

Attribute	Evidence from our project
1. Clear objective	Deliver cap, fill and label inspection with an operator interface by 30 September 2026 within ₹1,74,500 (Section 3.1).
2. Interdependent activities	The production pipeline (O) could not start until the student model (N) was accepted; Station 2 integration (R) needed both deployment (P) and the workstations (Q).
3. Various resources	Three students, a faculty guide, the college RTX A5000 GPU machine, open-source datasets and the Sathwik Solutions label dataset.
4. Specific time frame	173 working days planned, 2 February to 30 September 2026.
5. Unique, one-time effort	A one-off combination of a distilled YOLOv11 cap model, meniscus-based fill measurement and per-region label anomaly models.
6. Customer	Academic: the department and guide accept the work. Plant QA operators are the intended end users; no plant commissioned it.
7. Degree of uncertainty	Model quality could not be known before training. The label pipeline went through PatchCore, YOLO-based localisation and finally template matching with EfficientAD.

Table 2.1 - Project attributes evidenced

2.2 Balancing Project Constraints

Constraint	Priority	Target / limit	Justification
Quality	1	mAP50 $\geq$ 0.80, high-risk recall $\geq$ 0.85	A missed defect defeats the purpose of an inspection system.
Scope	2	Three inspection lines, operator UI, audit trail	Reviewers penalised claims beyond what was tested, so scope had to match evidence.
Schedule	3	30 September 2026	Academic deadlines are fixed, but internal dates had some room.
Resources	4	Three students; one shared college GPU	One shared machine carried all GPU work.
Risks	5	Discussed at guide reviews	Model outcomes were uncertain.
Customer satisfaction	6	Guide and department acceptance	Tested at guide reviews.
Stakeholder support	7	Guide, college, Sathwik Solutions	Needed for GPU access and label data.
Budget (₹)	8	₹1,91,950 notional; ₹0 cash	No cash at stake, so this ranked last.

Table 2.2 - Constraint prioritisation

Trade-off made (scope against schedule): once the cap and fill models ran late, we dropped cans, wrappers, boxes, QR validation and severity grading rather than move the date for all of them. We also accepted schedule against quality on the final build: 18 days of slip rather than releasing the TensorRT deployment with its batch mismatch unresolved.

2.3 Project Identification and Selection

Need. Thesis research found beverage inspection to be a small share of food-quality machine-learning work, and manual inspectors tire on fast lines. The literature survey of 2-16 February produced the references used in our thesis and paper. We did not measure line data ourselves, so the need is supported by literature rather than plant figures.

Selection. The options below are the three scopes visible in the project's records. The scoring (out of 10) is a retrospective formalisation of the choice we made.

Criterion	Weight (%)	A: Multi-product platform	B: Cap + fill + label	C: Cap inspector only
Feasible in the time available	30	3	7	9
Data availability	25	4	8	9
Technical and academic value	20	8	8	4
Industry relevance	15	9	8	5
Fits one college GPU	10	5	8	9
Weighted score	100	**5.35**	**7.70**	**7.40**

Table 2.3 - Project selection scoring (for B: 0.30×7 + 0.25×8 + 0.20×8 + 0.15×8 + 0.10×8 = 7.70)

Option	Key assumptions made	Data gathered to test them
A: Multi-product	Data exists for every package type; a 14-day sprint is enough	April plan's source list; only bottle data proved sufficient. CSITSS reviewers flagged the unvalidated claims (M3, C6)
B: Cap + fill + label	Open bottle datasets are enough; Sathwik Solutions data can support anomaly models	Cap dataset of 8,530 images; 80 good and 42 defective label images
C: Cap only	One defect type would satisfy the capstone	Would not cover fill or label failure modes identified in the survey

Table 2.4 - Assumptions behind each option

Option B won because A was not feasible on our data and time, while C scored close to B but had too little technical value and coverage for a capstone.

2.4 Stakeholder Engagement

Stakeholder	Interest in the project	Influence	How they were engaged
Nayana N Kumar (guide)	Academic quality, progress	H	Reviews about weekly, about 35 hours in all
Department of CSE (DS)	Assessment, GPU and facility	H	Formal thesis and presentation submissions
Sathwik Solutions	Industry tie-up, label data source	M	Provided the label dataset through our industry tie-up
CSITSS 2026 reviewers	Research rigour	M	Paper submission; three reviews drove our evidence and scoping decisions

Stakeholder	Interest in the project	Influence	How they were engaged
Plant QA operators	Usable verdicts, traceability	L	Not engaged directly; represented through operator-facing design choices
Project team	Learning, grades	H	Discussions among friends, role split by strengths, shared Git repository

Table 2.5 - Stakeholder register

2.5 Project Charter

Reconstructed in the form it could realistically have taken at project start; it was not signed at that time.

Field	Details
Title and purpose	VisionQAI. Replace tiring, inconsistent manual checks with traceable computer-vision inspection.
Project manager, authority	Sushanth S. Coordinates the plan and integration; scope changes are agreed by the team and raised with the guide.
Measurable objective	Three inspection lines with operator UI and audit trail by 30 September 2026, within ₹1,74,500.
Scope and out of scope	In: cap, fill and label inspection, dashboards, audit data, deployment. Out: live camera integration, physical rejection hardware, plant trials.
Deliverables and acceptance	Trained models (mAP50 $\geq$ 0.80), running pipeline with PASS, REJECT and UNCERTAIN verdicts, documentation, thesis. Accepted by the guide.
Milestones	Survey 13 Feb; scaffold 20 Mar; first demo 8 May; thesis 22 May; label models 7 Aug; Station 1 4 Sep; handover 30 Sep.
Budget and funding limit	TBC ₹1,74,500 plus reserve ₹17,450, limit ₹1,91,950, all notional. Cash ₹0.
Assumptions	College GPU stays available; public datasets are adequate; industry label data arrives.
Constraints	Three students alongside coursework; one GPU; no purchase budget.
Major risks	Model quality on scarce data; deployment on target GPU; scope creep.
Approval and success criteria	Guide approves milestones. Success: acceptance criteria in Table 3.1 met.
PM signature and date	_____ (Sushanth S), date: _____

Table 2.6 - Project charter

2.6 Request for Proposal and Soliciting Proposals

Not applicable. Nothing was bought in: the GPU and facility came from the college, and data came from open sources and our industry tie-up with Sathwik Solutions.

2.7 Life Cycle and Planning Process

Phase	Our dates	What happened in this phase
Initiating	2 Feb - 6 Mar	Literature survey, need identified, scope options weighed

Phase	Our dates	What happened in this phase
Planning	17 Feb - 6 Mar, revised April and August	Architecture and requirements; April plans; August resubmission plan
Performing	9 Mar - 26 Oct	Platform, models, label pipeline, final system, deployment
Closing	19 - 26 Oct, then this report	GPU verification, handover, lessons learned

Table 2.7 - The project on its life cycle

Planning steps and where they appear: objective (3.1); scope (3.2); WBS (3.4); responsibility (3.5); activities (4.1); sequence and network (4.2, Appendix B); resources and durations (4.1); costs (5.1); schedule (4.2, 4.4); budget (5.2); risks (6).

3. Project Planning

3.1 Project Objective

Deliver the VisionQAI inspection system, comprising cap, fill-level and label anomaly inspection with operator dashboards and an audit trail, by 30 September 2026, within a total budgeted cost of ₹1,74,500 notional and ₹0 cash. The expected benefit is consistent, traceable inspection decisions that can replace fatigue-prone manual checks and supply reviewed data for retraining.

3.2 Project Scope Document

Section	Our project
1. Customer requirements	Detect cap, fill and label defects from images; give a verdict with evidence; flag uncertain cases rather than guess; keep inspection records for review.
2. Statement of work	Survey the literature; prepare data; train and evaluate models; build backend and dashboards; integrate; package for deployment; verify on GPU; document.
3. Deliverables	Student cap model; two fill detectors; four EfficientAD label models; FastAPI backend and React dashboard; separate label dashboard; Docker and standalone launch scripts; thesis, paper, guides.
4. Acceptance criteria	Detector mAP50 $\geq$ 0.80 and high-risk recall $\geq$ 0.85; label-region recall $\geq$ 0.90; underfill below 0.40 and overfill above 0.85 of bottle height flagged; zero database write failures in stress test; both stations run and swap on the target GPU.
5. WBS	Table 3.3 and Appendix A.
Out of scope	Cans, wrappers and boxes; QR validation; S1-S4 severity grading; MongoDB and cloud hosting; live camera integration; physical ejection hardware; plant trials. Label inspection stays on its own dashboard by design.
Assumptions and constraints	College GPU available; data from open sources and Sathwik Solutions; only one set of models in GPU memory at a time.

Table 3.1 - Project scope document

3.3 Plan for Quality

Quality was checked from the first training runs, not only at the end. Results as recorded are given in Appendix E.

Quality measure	Target	How measured	When checked
Cap model mAP50	$\geq$ 0.80	Validation metrics in results.csv	Every training run
High-risk class recall	$\geq$ 0.85	Validation recall	Every training run
Label region recall	$\geq$ 0.90	Crop-level test scripts	After each calibration
Pipeline correctness	All scripts pass	test_step1-6, test_fill_step1-6, test_system_e2e	After each layer

Quality measure	Target	How measured	When checked
Audit integrity	0 write failures	stress_test.py	Before each release
GPU behaviour	Both stations and swap work	Demonstration on target GPU	Before handover

Table 3.2 - Quality plan

3.4 Work Breakdown Structure

WBS number	Deliverable / work package	Owner (one person)	TBC (₹)
1	**Initiation and research**		**₹19,000**
1.1	Literature survey (A)	Sushanth	₹10,600
1.2	Requirements and architecture (B)	Sushanth	₹8,400
2	**Inspection platform**		**₹24,650**
2.1	Platform scaffold and core (C)	Shreyas	₹6,200
2.2	Backend API and database (D)	Sushanth	₹9,050
2.3	Dashboard v1 and redesign (E)	Shreyas	₹9,400
3	**Cap and fill models (first system)**		**₹19,400**
3.1	Cap dataset, detector and classifier iterations (F)	Ullas	₹10,400
3.2	Fill dataset and fill-level models (G)	Ullas	₹5,600
3.3	First integrated demo (H)	Ullas	₹3,400
4	**Label anomaly pipeline**		**₹28,600**
4.1	Label: PatchCore baseline (J)	Ullas	₹6,400
4.2	Label: annotation, YOLO iteration, template matching (K)	Shreyas	₹9,000
4.3	Label: EfficientAD, grayscale, calibration (L)	Ullas	₹8,200
4.4	Label dashboard (M)	Shreyas	₹5,000
5	**Final system and deployment**		**₹40,250**
5.1	Cap teacher-student distillation (N)	Sushanth	₹7,050
5.2	Station 1 production pipeline (O)	Sushanth	₹10,050
5.3	Docker and TensorRT deployment (P)	Sushanth	₹7,250
5.4	Audit, Replay, Active Learning workstations (Q)	Sushanth	₹7,250
5.5	Station 2 integration, env-driven paths (R)	Sushanth	₹5,250
5.6	GPU verification and handover (S)	Ullas	₹3,400
6	**Reporting**		**₹10,600**
6.1	Thesis, IEEE paper, presentation (I)	Sushanth	₹10,600
7	**Guide reviews**		**₹32,000**
7.1	Weekly guide reviews (level of effort)	Sushanth	₹32,000
	Project total budgeted cost (TBC)		**₹1,74,500**

Table 3.3 - Work breakdown structure with work package budgets

3.5 Responsibility Assignment Matrix

Our team has three members, so the Member 4 column is omitted.

WBS work package	Sushanth	Ullas	Shreyas	Mentor	Client
1.1 Literature survey	P	S	S	S	
1.2 Requirements and architecture	P	S	S	S	S
2.1 Platform scaffold and core	S		P		
2.2 Backend API and database	P		S		
2.3 Dashboard v1 and redesign	S	S	P		
3.1 Cap dataset, detector and classifier iterations		P	S	S	
3.2 Fill dataset and fill-level models		P	S		
3.3 First integrated demo	S	P	S		
4.1 Label: PatchCore baseline		P	S	S	
4.2 Label: annotation, YOLO iteration, template matching		S	P		
4.3 Label: EfficientAD, grayscale, calibration	S	P		S	
4.4 Label dashboard			P		
5.1 Cap teacher-student distillation	P	S		S	
5.2 Station 1 production pipeline	P		S		
5.3 Docker and TensorRT deployment	P				
5.4 Audit, Replay, Active Learning workstations	P		S		
5.5 Station 2 integration, env-driven paths	P				
5.6 GPU verification and handover	S	P	S	S	S
6.1 Thesis, IEEE paper, presentation	P	S	S	S	S
7.1 Weekly guide reviews	P	S	S	S	

Table 3.4 - Responsibility assignment matrix (P = primary, S = support)

4. Developing the Schedule

4.1 Activity List, Resources and Durations

Method. The primary owner of each activity estimated it, drawing on what the work actually involved, so these are retrospective estimates from commit history and team recall, not forecasts made at the time. Hours are shown by member (Su, Ul, Sh). Guide reviews (WBS 7.1) are level of effort and are not in the network.

#	Activity	Predecessors	Resources (hours)	Duration (days)	Estimated by
A	Literature survey	-	Su 16, Ul 16, Sh 17	10	Sushanth
B	Requirements and architecture	A	Su 20, Ul 8, Sh 9	15	Sushanth
C	Platform scaffold and core	B	Sh 21, Su 8	10	Shreyas
D	Backend API and database	C	Su 33, Sh 4	20	Sushanth

#	Activity	Predecessors	Resources (hours)	Duration (days)	Estimated by
E	Dashboard v1 and redesign	C	Sh 38, Su 4, Ul 4	30	Shreyas
F	Cap dataset, detector and classifier iterations	B	Ul 44, Sh 8	25	Ullas
G	Fill dataset and fill-level models	F	Ul 20, Sh 8	10	Ullas
H	First integrated demo	D, E, G	Ul 8, Su 4, Sh 4	5	Ullas
I	Thesis, IEEE paper, presentation	H	Su 20, Ul 20, Sh 8	10	Sushanth
J	Label: PatchCore baseline	I	Ul 28, Sh 4	25	Ullas
K	Label: annotation, YOLO iteration, template matching	J	Sh 25, Ul 20	15	Shreyas
L	Label: EfficientAD, grayscale, calibration	K	Ul 36, Su 4	15	Ullas
M	Label dashboard	L	Sh 25	15	Shreyas
N	Cap teacher-student distillation	L	Su 25, Ul 4	10	Sushanth
O	Station 1 production pipeline	N	Su 37, Sh 4	10	Sushanth
P	Docker and TensorRT deployment	O	Su 29	8	Sushanth
Q	Audit, Replay, Active Learning workstations	O	Su 29	10	Sushanth
R	Station 2 integration, env-driven paths	P, Q	Su 21	5	Sushanth
S	GPU verification and handover	M, R	Ul 8, Su 4, Sh 4	3	Ullas

Table 4.1 - Activity list with resource and duration estimates

4.2 Network Diagram and Schedule Calculations

The network diagram is in Appendix B. Days are working days counted from 0 at the start of 2 February 2026.

#	Activity	Dur.	ES	EF	LS	LF	Total slack	Free slack	Critical?
A	Literature survey	10	0	10	0	10	0	0	**Yes**
B	Requirements and architecture	15	10	25	10	25	0	0	**Yes**
C	Platform scaffold and core	10	25	35	25	35	0	0	**Yes**
D	Backend API and database	20	35	55	45	65	10	10	No
E	Dashboard v1 and redesign	30	35	65	35	65	0	0	**Yes**
F	Cap dataset, detector and classifier iterations	25	25	50	30	55	5	0	No
G	Fill dataset and fill-level models	10	50	60	55	65	5	5	No
H	First integrated demo	5	65	70	65	70	0	0	**Yes**
I	Thesis, IEEE paper, presentation	10	70	80	70	80	0	0	**Yes**
J	Label: PatchCore baseline	25	80	105	80	105	0	0	**Yes**
K	Label: annotation, YOLO iteration, template matching	15	105	120	105	120	0	0	**Yes**
L	Label: EfficientAD, grayscale, calibration	15	120	135	120	135	0	0	**Yes**

#	Activity	Dur.	ES	EF	LS	LF	Total slack	Free slack	Critical?
M	Label dashboard	15	135	150	155	170	20	20	No
N	Cap teacher-student distillation	10	135	145	135	145	0	0	**Yes**
O	Station 1 production pipeline	10	145	155	145	155	0	0	**Yes**
P	Docker and TensorRT deployment	8	155	163	157	165	2	2	No
Q	Audit, Replay, Active Learning workstations	10	155	165	155	165	0	0	**Yes**
R	Station 2 integration, env-driven paths	5	165	170	165	170	0	0	**Yes**
S	GPU verification and handover	3	170	173	170	173	0	0	**Yes**

Table 4.2 - Forward and backward pass, slack and critical path

Critical path: A – B – C – E – H – I – J – K – L – N – O – Q – R – S, totalling 173 working days.

Worked calculation for activity D (Backend API and database, 20 days).

- Forward pass, Rule 1: D's only predecessor is C, with EF 35, so ES = 35 and EF = 35 + 20 = 55.
- Backward pass, Rule 2: D's only successor is H, with LS 65, so LF = 65 and LS = 65 - 20 = 45.
- Total slack = LS - ES = 45 - 35 = 10 days.
- Free slack = earliest successor ES - EF = 65 - 55 = 10 days.

(a) Activity M, Label dashboard, has the most total slack at 20 days. We would use it to lend Shreyas to critical-path work when behind, and indeed the dashboard consumed 15 of those days without moving the finish.

(b) The largest critical activity is E, Dashboard v1 and redesign, at 30 days. A five-day slip would move the finish from day 173 to day 178, 7 October 2026, making the total slack on the critical path -5 against the required date. Paths joining E at H gain slack against the later finish: D rises from 10 to 15 days, and F and G from 5 to 10. Paths after H keep theirs, because their dates all move together: M stays at 20 days and P at 2.

4.3 Milestones

Milestone	Planned date	Actual date	Status
Literature survey complete	13 Feb	16 Feb	Late by 1 day
Requirements and architecture complete	6 Mar	6 Mar	Met
Platform scaffold pushed	20 Mar	21 Mar	Met
Cap models accepted	10 Apr	3 May	Late by 15 days
First integrated demo	8 May	15 May	Late by 5 days
Thesis complete	22 May	22 May	Met
Label models calibrated	7 Aug	7 Aug	Met
Student cap model accepted	21 Aug	7 Sep	Late by 11 days
Station 1 production pipeline	4 Sep	14 Sep	Late by 6 days

Milestone	Planned date	Actual date	Status
Station 2 integrated	25 Sep	26 Oct	Late by 21 days
GPU verification and handover	30 Sep	26 Oct	Late by 18 days

Table 4.3 - Milestones, all in 2026 (a milestone has zero duration)

4.4 Baseline Schedule

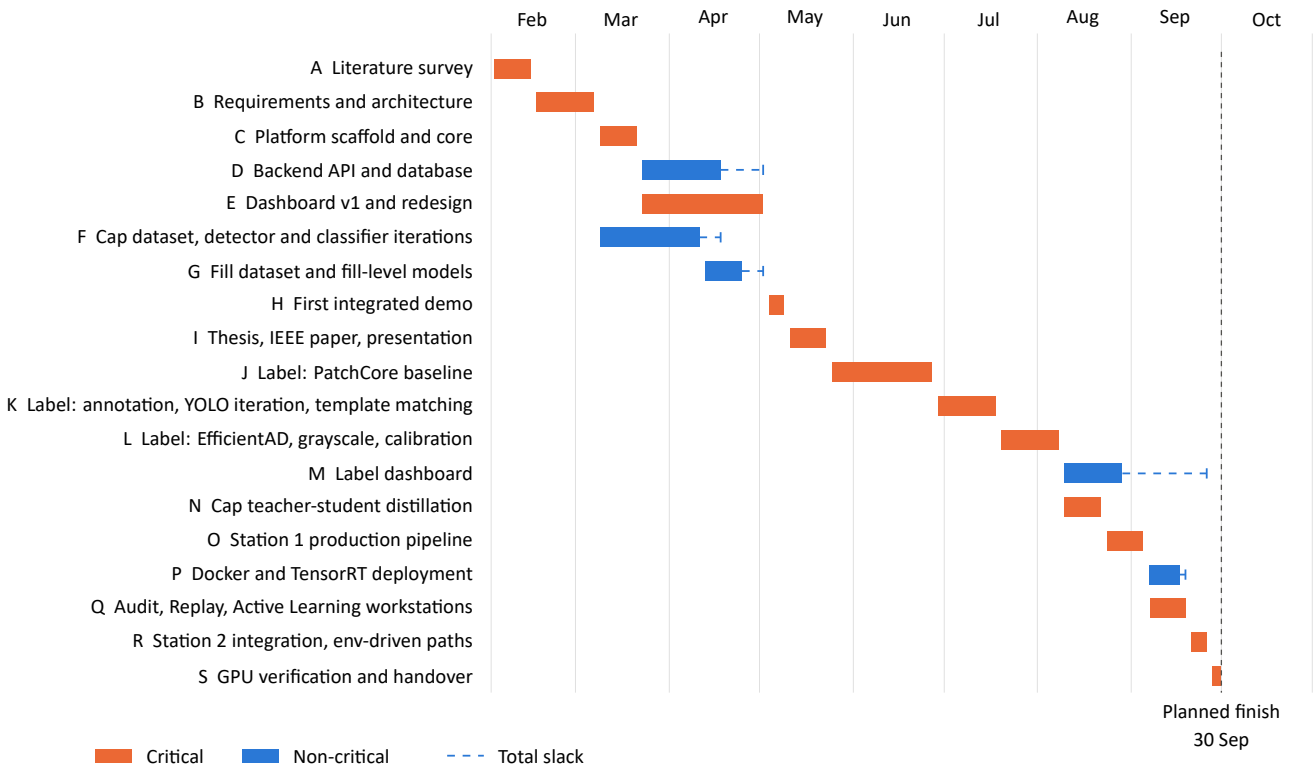


Figure 4.1 - Baseline schedule, critical path in orange, total slack dashed

Planned finish and required completion date are both 30 September 2026, so total slack on the critical path is zero.

4.5 Project Control Process

The reporting period is one calendar month, matched to how the work actually ran alongside course-work; guide reviews took place about weekly. Data for each period are actual start and finish (from commits), hours, costs expended and committed (always ₹0 committed) and percent complete. The team and the guide attend reviews. The threshold we set when formalising this process is a slip of more than 5 working days on a critical activity, or a CPI below 0.90.

We did not run this process formally at the time: the baseline was formalised retrospectively for this report, and the monthly data in Appendix C are reconstructed from Git. Real control happened through discussion among the team and the guide's reviews.

4.6 Effects of Actual Schedule Performance

Activities not listed finished on their planned date.

Activity	Planned finish	Actual finish	Variance (days)	Cause	Effect on total slack
A Literature survey	13 Feb	16 Feb	+1	Survey ran one day past its two weeks	-1 on the critical path; recovered within B
E Dashboard v1 and redesign	1 May	4 May	+1	Redesign landed on 4 May	-1 on the critical path
F Cap dataset, detector and classifier iterations	10 Apr	3 May	+15	Bottle cap models needed repeated iteration: small caps, class mapping, white-only colour bias	Used all 5 days of slack; path at -10
G Fill dataset and fill-level models	24 Apr	15 May	+15	Inherited F's delay; water datasets had to be merged	Path stayed at -10
H First integrated demo	8 May	15 May	+5	Waited on G	-5 on the critical path; recovered by overlapping I
J Label: PatchCore baseline	26 Jun	11 Jul	+10	Paper revision and a pause after the thesis delayed the start to mid-June	-10 on the critical path
K Label: annotation, YOLO iteration, template matching	17 Jul	4 Aug	+12	YOLO ROI detector tried, then replaced by template matching	-12 on the critical path
M Label dashboard	28 Aug	18 Sep	+15	Label dashboard finished after the models	Used 15 of its 20 days of slack
N Cap teacher-student distillation	21 Aug	7 Sep	+11	CSITSS rejection (22 Aug) took priority, delaying the start	-11 on the critical path
O Station 1 production pipeline	4 Sep	14 Sep	+6	Inherited N's delay	-6 on the critical path
P Docker and TensorRT deployment	16 Sep	22 Oct	+26	TensorRT static batch mismatch; dynamic batching and versioning reworked	Used its 2 days of slack; path at -24
Q Audit, Replay, Active Learning workstations	18 Sep	22 Oct	+24	Started after the rework on P began; delivered with P on 22 Oct	-24 on the critical path
R Station 2 integration, env-driven paths	25 Sep	26 Oct	+21	Waited on P and Q; Station 2 merge into one UI	-21 on the critical path
S GPU verification and handover	30 Sep	26 Oct	+18	Waited on R	Project finished 18 days late

Table 4.4 - Schedule variance analysis

The early phase recovered by overlapping the thesis with the demo work, so I finished on 22 May as planned, and L also recovered by overlapping K. The later slips did not recover because the distillation and deployment activities depend strictly on each other.

4.7 Incorporating Changes and Updating the Schedule

No change forms were used at the time; this log records the decisions visible in the repository, with impact estimated retrospectively.

ID	Description	Impact (scope / days / ₹)	Decision	Approved by
CR1	Narrow from multi-product to bottle and label lines	Scope reduced; avoided work outside the baseline	Approved	Team, guide
CR2	Replace PatchCore with per-region EfficientAD	No scope change; part of 10-day J slip; +₹1,600	Approved	Team
CR3	Replace YOLO localisation with template matching	No scope change; 12-day K slip; +₹2,600	Approved	Team
CR4	Keep label inspection on its own dashboard	Removes merge work; no date impact	Approved	Team
CR5	Distil YOLOv11x teacher into YOLOv11m student	Inside N; 11-day slip; +₹500	Approved	Team
CR6	Rework TensorRT to dynamic batching and versioned engines	Quality up; 26-day P slip; +₹1,500	Approved	PM

Table 4.5 - Change request log

Updated schedule. The final finish date is 26 October 2026, 18 working days after the baseline. The revised critical path in the final phase ran through P and Q (both finishing 22 October), then R, ending with S on 26 October. Measured against the 30 September requirement, total slack on that path ended at -18 days.

5. Determining Cost and Budget

5.1 Estimate Activity Costs and the Total Budgeted Cost

Every estimate is a retrospective estimate from commit history and team recall: hours from Table 4.1 multiplied by the rate card (Table 0.3). Materials, equipment, facilities, travel and subcontractors are ₹0, as the college provided the GPU and facility without charge and the data cost nothing. The GPU use is therefore uncharged rather than free of value. The reserve is held outside the TBC, so earned value is measured only against planned work.

WBS num-ber	Work package	Labour (₹)	Materials / other (₹)	Basis of estimate	TBC (₹)
1.1	Literature survey	₹10,600	₹0	3 members for 2 weeks at about 8 h/week	₹10,600
1.2	Requirements and architecture	₹8,400	₹0	Mostly PM; sized against the 15-day duration	₹8,400
2.1	Platform scaffold and core	₹6,200	₹0	Size of the Phase 0-4 scaffold	₹6,200
2.2	Backend API and database	₹9,050	₹0	API routers and DB schema count	₹9,050

WBS number	Work package	Labour (₹)	Materials / other (₹)	Basis of estimate	TBC (₹)
2.3	Dashboard v1 and redesign	₹9,400	₹0	Page count of the dashboard	₹9,400
3.1	Cap dataset, detector and classifier iterations	₹10,400	₹0	Training runs expected, plus analysis	₹10,400
3.2	Fill dataset and fill-level models	₹5,600	₹0	Two detectors and fill-ratio logic	₹5,600
3.3	First integrated demo	₹3,400	₹0	Integration week	₹3,400
4.1	Label: PatchCore baseline	₹6,400	₹0	One anomaly baseline	₹6,400
4.2	Label: annotation, YOLO iteration, template matching	₹9,000	₹0	528 crops plus localisation trials	₹9,000
4.3	Label: EfficientAD, grayscale, calibration	₹8,200	₹0	4 ROI models with calibration	₹8,200
4.4	Label dashboard	₹5,000	₹0	Single-purpose dashboard	₹5,000
5.1	Cap teacher-student distillation	₹7,050	₹0	Teacher and student training watch	₹7,050
5.2	Station 1 production pipeline	₹10,050	₹0	Six pipeline layers	₹10,050
5.3	Docker and TensorRT deployment	₹7,250	₹0	Compose, engine compile, stress test	₹7,250
5.4	Audit, Replay, Active Learning workstations	₹7,250	₹0	Three operator pages	₹7,250
5.5	Station 2 integration, env-driven paths	₹5,250	₹0	Mirror of Station 1 layout	₹5,250
5.6	GPU verification and handover	₹3,400	₹0	Checks on target PC and handover	₹3,400
6.1	Thesis, IEEE paper, presentation	₹10,600	₹0	Thesis, paper and slides, two weeks	₹10,600
7.1	Weekly guide reviews	₹32,000	₹0	32 h planned at ₹1,000/h	₹32,000
	Total budgeted cost (TBC)	**₹1,74,500**	**₹0**	**Sum of work packages**	**₹1,74,500**
	Contingency reserve (held separately)			**10% of TBC**	**₹17,450**
	Funding limit			**TBC + reserve**	**₹1,91,950**

Table 5.1 - Activity cost estimates and total budgeted cost (all amounts in ₹)

5.2 Time-Phased Budget (Cumulative Budgeted Cost)

Each package is spread evenly over its planned working days. Rows use activity letters from Table 4.1. Amounts are in ₹.

WP	P1 Feb	P2 Mar	P3 Apr	P4 May	P5 Jun	P6 Jul	P7 Aug	P8 Sep	P9 Oct	Total (₹)
A	10,600									10,600
B	5,600	2,800								8,400
C		6,200								6,200

WP	P1 Feb	P2 Mar	P3 Apr	P4 May	P5 Jun	P6 Jul	P7 Aug	P8 Sep	P9 Oct	Total (₹)
D		3,168	5,882							9,050
E		2,193	6,894	313						9,400
F		7,072	3,328							10,400
G			5,600							5,600
H				3,400						3,400
I				10,600						10,600
J				1,280	5,120					6,400
K					1,200	7,800				9,000
L						5,467	2,733			8,200
M							5,000			5,000
N							7,050			7,050
O							6,030	4,020		10,050
P								7,250		7,250
Q								7,250		7,250
R								5,250		5,250
S								3,400		3,400
7.1	3,699	4,070	4,069	3,885	4,069	4,254	3,885	4,069		32,000
Period	**19,899**	**25,503**	**25,773**	**19,478**	**10,389**	**17,521**	**24,698**	**31,239**	**0**	**174,500**
CBC	**19,899**	**45,402**	**71,175**	**90,653**	**101,042**	**118,563**	**143,261**	**174,500**	**174,500**	

Table 5.2 - Time-phased budget; the last row is the cumulative budgeted cost (CBC)

5.3 Actual Cost and Committed Cost

Spent means hours from the reconstruction valued at the rate card. No cash was paid and nothing was ordered, so committed cost is ₹0 throughout.

WBS	Work package	CBC to date (₹)	Spent (₹, notional)	Committed (₹)	CAC (₹)
1.1	Literature survey	₹10,600	₹10,600	₹0	₹10,600
1.2	Requirements and architecture	₹8,400	₹7,900	₹0	₹7,900
2.1	Platform scaffold and core	₹6,200	₹6,200	₹0	₹6,200
2.2	Backend API and database	₹9,050	₹8,850	₹0	₹8,850
2.3	Dashboard v1 and redesign	₹9,400	₹11,000	₹0	₹11,000
3.1	Cap dataset, detector and classifier iterations	₹10,400	₹12,000	₹0	₹12,000
3.2	Fill dataset and fill-level models	₹5,600	₹6,600	₹0	₹6,600
3.3	First integrated demo	₹3,400	₹3,400	₹0	₹3,400
4.1	Label: PatchCore baseline	₹6,400	₹8,000	₹0	₹8,000
4.2	Label: annotation, YOLO iteration, template matching	₹9,000	₹11,600	₹0	₹11,600
4.3	Label: EfficientAD, grayscale, calibration	₹8,200	₹8,200	₹0	₹8,200
4.4	Label dashboard	₹5,000	₹6,800	₹0	₹6,800

WBS	Work package	CBC to date (₹)	Spent (₹, notional)	Committed (₹)	CAC (₹)
5.1	Cap teacher-student distillation	₹7,050	₹7,550	₹0	₹7,550
5.2	Station 1 production pipeline	₹10,050	₹10,050	₹0	₹10,050
5.3	Docker and TensorRT deployment	₹7,250	₹8,750	₹0	₹8,750
5.4	Audit, Replay, Active Learning workstations	₹7,250	₹7,850	₹0	₹7,850
5.5	Station 2 integration, env-driven paths	₹5,250	₹5,250	₹0	₹5,250
5.6	GPU verification and handover	₹3,400	₹4,200	₹0	₹4,200
6.1	Thesis, IEEE paper, presentation	₹10,600	₹9,700	₹0	₹9,700
7.1	Weekly guide reviews	₹32,000	₹35,000	₹0	₹35,000
	Total	**₹1,74,500**	**₹1,89,500**	**₹0**	**₹1,89,500**

Table 5.3 - Actual and committed cost as of the Status Date

5.4 Earned Value of the Work Performed

Percent complete at the Status Date was judged by Sushanth S as PM, against an artifact that can be checked in the repository, not by opinion. For monthly data, an in-progress package is credited with the elapsed share of its actual window, rounded to the nearest 5% and held below 100% until its finishing commit.

WBS	Work package	TBC (₹)	Percent complete	How judged	CEV (₹)
1.1	Literature survey	₹10,600	100%	Reference list in the thesis	₹10,600
1.2	Requirements and architecture	₹8,400	100%	Planning documents exist	₹8,400
2.1	Platform scaffold and core	₹6,200	100%	Code in repo	₹6,200
2.2	Backend API and database	₹9,050	100%	API and schema commits	₹9,050
2.3	Dashboard v1 and redesign	₹9,400	100%	Dashboard commits	₹9,400
3.1	Cap dataset, detector and classifier iterations	₹10,400	100%	Models in repo, metrics logged	₹10,400
3.2	Fill dataset and fill-level models	₹5,600	100%	Fill models run in pipeline	₹5,600
3.3	First integrated demo	₹3,400	100%	Commit 8c54630	₹3,400
4.1	Label: PatchCore baseline	₹6,400	100%	PatchCore scripts, membank	₹6,400
4.2	Label: annotation, YOLO iteration, template matching	₹9,000	100%	Crops, localisation committed	₹9,000
4.3	Label: EfficientAD, grayscale, calibration	₹8,200	100%	Thresholds, crop test records	₹8,200
4.4	Label dashboard	₹5,000	100%	Separate dashboard runs	₹5,000
5.1	Cap teacher-student distillation	₹7,050	100%	results.csv, 126 epochs	₹7,050
5.2	Station 1 production pipeline	₹10,050	100%	test_step1-6 scripts	₹10,050
5.3	Docker and TensorRT deployment	₹7,250	100%	Compose files, stress test	₹7,250
5.4	Audit, Replay, Active Learning workstations	₹7,250	100%	Commit c8f0524	₹7,250

WBS	Work package	TBC (₹)	Percent complete	How judged	CEV (₹)
5.5	Station 2 integration, env-driven paths	₹5,250	100%	test_system_e2e.py	₹5,250
5.6	GPU verification and handover	₹3,400	100%	Team demonstration	₹3,400
6.1	Thesis, IEEE paper, presentation	₹10,600	100%	Thesis and plagiarism report	₹10,600
7.1	Weekly guide reviews	₹32,000	100%	Level of effort	₹32,000
	Total	**₹1,74,500**	**100%**		**₹1,74,500**

Table 5.4 - Cumulative earned value

5.5 Analyse Cost Performance

Measure	Formula	Value	What it tells us
TBC	Sum of work package budgets	₹1,74,500	Performance baseline
CBC	Budgeted for work scheduled to date	₹1,74,500	All work was due by 30 Sep
CAC	Spent plus committed	₹1,89,500 + ₹0 = ₹1,89,500	Notional effort used
CEV	TBC × percent complete	₹1,74,500 × 100% = ₹1,74,500	All planned work earned
CPI	CEV / CAC	₹1,74,500 / ₹1,89,500 = 0.921	₹0.921 of value per ₹1 of effort
CV	CEV - CAC	₹1,74,500 - ₹1,89,500 = -₹15,000	Over budget, within reserve

Table 5.5 - Cost performance as of the Status Date

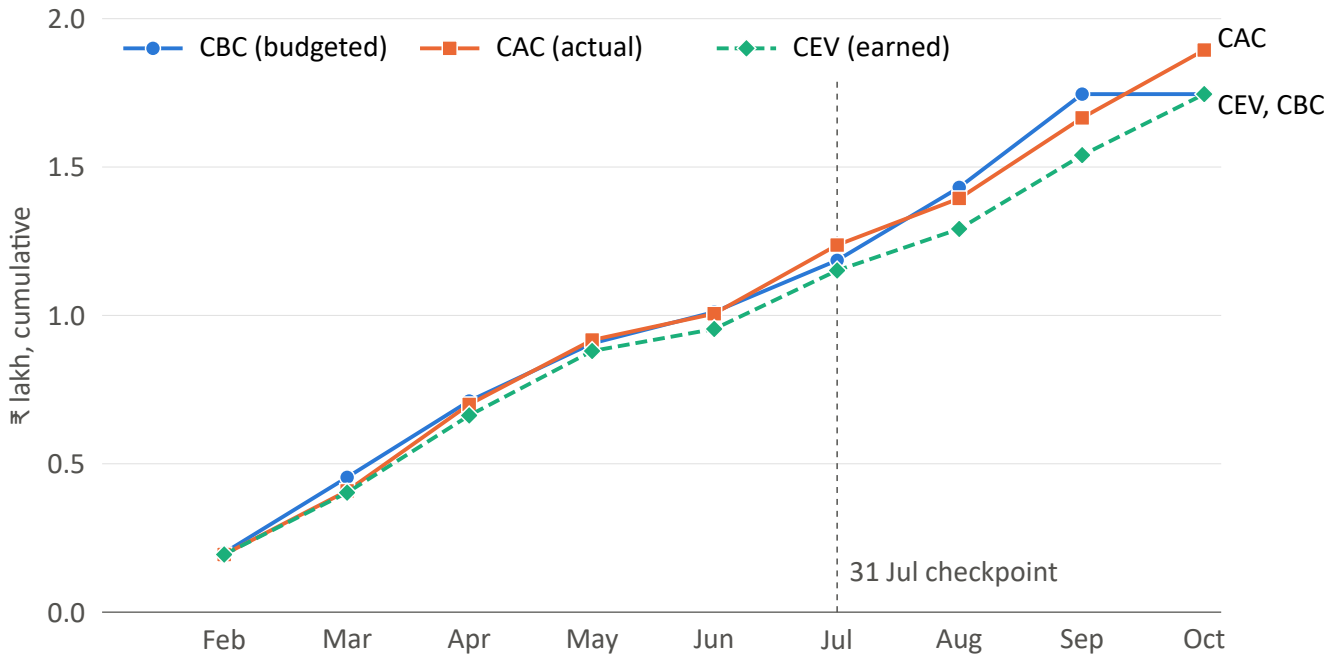


Figure 5.1 - Cumulative budgeted, actual and earned value curves. Table 5.8 gives the values.

What the picture shows. CAC is above CEV from March onward and the gap widens, so we were consistently paying slightly more for each unit of work. CEV falls well behind CBC during August and September, which is when the schedule slip appears as money: planned work was not being earned. In October CBC is flat because all work was due by then, and CEV catches up with it at 100%, but CAC does not come back down.

5.6 Forecast the Cost at Completion

At the Status Date the work is done, so both forecasts equal the final cost and the TCPI has no remaining work to apply to.

Measure	Formula	Value	Interpretation
FCAC, method 1	TBC / CPI	₹1,74,500 / 0.921 = ₹1,89,500	Equals CAC at completion
FCAC, method 2	CAC + (TBC - CEV)	₹1,89,500 + ₹0 = ₹1,89,500	Equals CAC at completion
Overrun or underrun	TBC - FCAC	₹1,74,500 - ₹1,89,500 = -₹15,000	Overrun; 86% of reserve used
TCPI	(TBC - CEV) / (TBC - CAC)	₹0 / -₹15,000 = 0	Finishing within TBC is no longer possible

Table 5.6 - Forecasted cost at completion

Interim forecast, 31 July 2026. A completed project cannot show a live forecast, so the table below applies the same methods at the last month before the deployment slips, when 66.1% of the value was earned.

Measure at 31 July	Working	Value	Reading
CBC, CAC, CEV	From Table 5.8	₹1,18,563, ₹1,23,774, ₹1,15,260	Slightly behind and over
CPI	₹1,15,260 / ₹1,23,774	0.931	Below 1
CV	₹1,15,260 - ₹1,23,774	-₹8,514	Over budget
FCAC, method 1	₹1,74,500 / 0.931	₹1,87,390	Rest at today's efficiency
FCAC, method 2	₹1,23,774 + (₹1,74,500 - ₹1,15,260)	₹1,83,014	Rest exactly to budget
TCPI	(₹1,74,500 - ₹1,15,260) / (₹1,74,500 - ₹1,23,774)	1.168	Above our CPI, so unlikely

Table 5.7 - Interim cost forecast, 31 July 2026

Method 1 was the more realistic assumption, because the overruns came from recurring causes, model iteration and deployment rework, not one-off events. The final cost of ₹1,89,500 was in fact above both forecasts, as the deployment rework was still to come. A TCPI above 1 meant that staying within the TBC would have needed better efficiency than we had achieved, which was unlikely, so the reserve was the realistic cover.

If CPI had fallen to 0.80: FCAC = ₹1,74,500 / 0.80 = ₹2,18,125, an overrun of ₹43,625, more than the ₹17,450 reserve. We would have told the PM to cut remaining scope, since extra funding has no meaning here and each extra rupee is really student hours.

Month end	CBC	CAC	CEV	CPI	CV	CEV - CBC
Feb	₹19,899	₹19,344	₹19,411	1.003	₹67	-₹488
Mar	₹45,402	₹40,778	₹40,354	0.990	-₹424	-₹5,048
Apr	₹71,175	₹70,138	₹66,303	0.945	-₹3,835	-₹4,872
May	₹90,653	₹91,826	₹87,891	0.957	-₹3,935	-₹2,762
Jun	₹1,01,042	₹1,00,657	₹95,417	0.948	-₹5,240	-₹5,625
Jul	₹1,18,563	₹1,23,774	₹1,15,260	0.931	-₹8,514	-₹3,303
Aug	₹1,43,261	₹1,39,465	₹1,29,176	0.926	-₹10,289	-₹14,085
Sep	₹1,74,500	₹1,66,519	₹1,54,134	0.926	-₹12,385	-₹20,366
Oct	₹1,74,500	₹1,89,500	₹1,74,500	0.921	-₹15,000	₹0

Table 5.8 - Monthly control data behind Figure 5.1 (October is the Status Date)

6. Managing Risk

6.1 Identify and Assess Risks

Risks are listed in rank order. The register was formalised for this report, based on what the project's records show.

ID	Risk	Category	Potential impact	Likeli-hood	Impact	On CP?
R1	Repeated model iteration on scarce or biased data	Technical	Weeks of slip; targets missed	H	H	Yes (J, K)
R2	TensorRT engines are GPU-specific and may fail on the target	Technical	Deployment rework; late finish	H	H	Partly (P, near)
R3	Scope creep from the multi-product vision	Customer	Unfinishable plan; overclaiming	H	M	Yes
R4	Paper or thesis rejected for lack of evidence	External	Rework; lost weeks	M	M	Yes (N)
R5	Label region ROI_4 stays noisy, with false rejects	Technical	Line stoppages from false alarms	M	M	Yes (L)
R6	Model files not in Git get lost or not copied	Schedule	Target machine cannot run	M	M	Yes (S)
R7	Two model sets exceed GPU memory	Technical	Crash on station switch	M	M	Yes (R)
R8	Member availability falls during exams or after submissions	Human res.	Idle weeks	M	L	Yes (J)

Table 6.1 - Risk identification and assessment

6.2 Plan Risk Responses

Contingency amounts add up to the ₹17,450 reserve in Table 5.1.

Risk	Response	Specific action	Action on trigger	Respon-sibility	Contin-gency (₹)
R1	Mitigate	Fix a baseline first; test on crops; change method if no gain	A run misses target twice	Ullas	₹6,410
R2	Mitigate	Fall back from engine to ONNX to PT; version engines	Warmup shape mismatch or failed compile	Sushanth	₹3,660
R3	Avoid	Bottle and label lines only; log scope in change log	A feature with no test data is proposed	Sushanth	₹0
R4	Mitigate	Claim only what was tested; prepare a resubmission plan	Reviewer rates work borderline	Sushanth	₹2,290
R5	Accept	Report honestly; re-annotate if time allows	False positives above 10 on a region	Ullas	₹2,330

Risk	Response	Specific action	Action on trigger	Respon-sibility	Contin-gency (₹)
R6	Mitigate	Manual-copy caution in the target guide	Missing file on new machine	Shreyas	₹920
R7	Avoid	Load one station at a time; lazy-load fill engine	VRAM not released on swap	Sushanth	₹920
R8	Accept	Reorder work among members	Member below 2 hours in a week	Sushanth	₹920

Table 6.2 - Risk assessment matrix: response plans

6.3 Monitor Risks

No written matrix existed during the project. Risks were discussed informally in team conversations and at the roughly weekly guide reviews. R5 and R7 were added during the project; the others follow from the original scope. Actual cost is the overrun on the affected packages from Table 5.3.

Risk	Occurred?	Did the trigger warn in time?	How the response performed	Actual cost (₹)
R1	Yes	Partly; the 21.4% end-to-end recall exposed stale weights only at test time	Switching to grayscale raised ROI 1-3 recall above 92%; methods changed twice	₹6,800
R2	Yes	Yes; the warmup mismatch showed it before release	Dynamic batching and versioned engines fixed it, at 26 days	₹1,500
R3	Yes	Late; the review confirmed it	Scope narrowed (Section 7.2); no later growth	₹0
R4	Yes	No warning before the 22 Aug result	Resubmission plan written; delayed N by 11 days	₹500
R5	Yes	Yes; crop test showed 56 false positives	Accepted and reported; remains a known weakness	₹0
R6	No	Not triggered	Guide caution in place	₹0
R7	No	Not triggered	One-station-at-a-time design held	₹0
R8	Partly	No	Gap between 22 May and 15 June absorbed in J's slip	₹0

Table 6.3 - Risk monitoring record

7. The Project Team

7.1 Team Effectiveness

Characteristic	Rating (1-5)	Evidence from our project
Clear understanding of the objective	3	The objective moved from a multi-product platform to three lines before it was stable (CR1)
Clear expectations of roles	4	Written role splits in April (Collaborator A and B plans) and a clean ownership pattern in the commits
Results orientation	5	Three inspection lines delivered at 100%, despite an 18-day slip
Cooperation and collaboration	4	Each person supported others' packages (Table 3.4); S was led by Ullas with both others supporting
High level of trust	4	Friends working by comfort, with no timesheets; the reliance on trust is also why we had no records

Table 7.1 - Team effectiveness

Biggest barrier: the lack of a shared written plan. We worked by talking with each other, which suited a team of friends but left no baseline, so slips were noticed by feel rather than measured. What we did about it was limited at the time; we introduced written role splits in April, and this report's reconstruction shows what we should have kept from the start.

7.2 Conflict and Problem Solving

The problem. The April plan asked for four package types, label anomaly detection, QR validation and S1-S4 severity grading. As the work progressed, the team had to decide what to keep. The **source** was work scope. We handled it by **collaborating**: we discussed it together and agreed on the narrower scope. This was the right approach, as each member held evidence the others needed (data availability, model results, integration effort), and a forced choice would have ignored some of it.

Step	What our team did
1. Problem statement	The plan covered 4 package types plus QR and severity in 14 days; working evidence existed for one package type at that time.
2. Potential causes	Scope set from an aspirational design; too little data per product; underestimated iteration; no measure of progress.
3. Data gathered, cause verified	Cap dataset 8,530 images against none adequate for cans or wrappers. CSITSS reviewers (22 Aug record) flagged multi-packaging claims as unvalidated (M3) and overclaiming (C6). Likely cause: scope set ahead of data.
4. Possible solutions	Keep the full scope and extend the date; build a shared platform with simulated products; narrow to lines with real data.
5. Criteria and evaluation	Testable with real data, finishable by the deadline, still a substantial capstone. Only narrowing met all three.
6. Best solution	Bottle cap, fill level and label anomaly lines only.

Step	What our team did
7. Plan revised	Removed cans, wrappers, boxes, QR, severity grading and MongoDB; recorded as CR1 and in the out-of-scope list.
8. Implemented	Later work followed the narrower scope: distillation, Station 1, Station 2, and a separate label dashboard.
9. Checked against step 1	Yes for the scope measure: all three retained lines have working models and evidence. The cost was accepted: the system is no longer multi-product.

Table 7.2 - Problem-solving record

8. Lessons Learned and Conclusion

What worked. Clear ownership: each package had one owner, and work rarely collided (Table 3.4). Trying a baseline before committing: the label line moved to EfficientAD on evidence from crop tests, not preference (CR2, CR3). Building a stable test script per layer let the late Station 2 merge land without disturbing Station 1.

What we would do differently. Keep a simple weekly log from day one; this report had to reconstruct what ten minutes a week would have recorded. Set scope from available data, not ambition (Section 7.2). Test deployment on the target GPU much earlier: P's 26-day slip (Table 4.4) came from issues that an early trial could have found while the path still had time.

Course tool or concept	How we applied it	Useful (1-5)	What we would do differently
Project charter	Reconstructed (Table 2.6)	3	Write it at the start; it would have exposed the scope problem in February
Work breakdown structure	19 packages with one owner each	5	Use it to drive task assignment in Git
Responsibility matrix	One P per row (Table 3.4)	4	Agree it openly, rather than letting it emerge
Network diagram and slack	Table 4.2 and Appendix B	4	Would have shown that deployment had only 2 days of slack
Time-phased budget and earned value	Tables 5.2-5.8	3	With ₹0 cash, useful mainly as a measure of hours; needs real time data
Risk assessment matrix	Tables 6.1-6.3	4	Keep it live; the triggers only help if someone watches them

Table 8.1 - Reflection on the tools applied

For a three-person student project, the WBS and network earned their keep most clearly. Earned value earned less, as cash was nil and hours were not tracked.

Outcome against the objective. Deliverable: three inspection lines with dashboards and audit trail delivered, with the multi-product vision deliberately removed. Schedule: finished 26 October against

30 September, 18 working days late. Budget: ₹1,89,500 notional against a TBC of ₹1,74,500, CPI 0.921, within the ₹1,91,950 funding limit, and ₹0 cash.

9. Individual Contribution and Reflection

Member	WBS packages owned (P)	Hours worked	Main contribution	Reflection attached?
Sushanth S	1.1, 1.2, 2.2, 5.1, 5.2, 5.3, 5.4, 5.5, 6.1, 7.1	250	PM; backend, distillation, production pipeline, Docker and TensorRT, final UI, Station 2 merge	Yes (below)
Ullas N	3.1, 3.2, 3.3, 4.1, 4.3, 5.6	240	Cap and fill model iterations, label anomaly models, GPU verification and handover	Yes (below)
Shreyas Bairy K S	2.1, 2.3, 4.2, 4.4	220	Platform scaffold, dashboards, template matching for label localisation	Yes (below)

Table 9.1 - Individual contribution

Sushanth S (Project Manager)

I owned the literature survey, requirements, backend, thesis coordination, distillation, production pipeline, Docker and TensorRT deployment, the operator workstations and the Station 2 merge, 250 hours in total. As PM I also coordinated the plan and kept integration moving between the three of us. The work that taught me most was deployment, where static batch sizes in the compiled engine broke multi-camera inference. Dynamic batching and versioned engines gave us a sounder system, but the problem showed me how one technical assumption can hold up everything that depends on it.

One thing I got wrong, and what I learned about project management from it: I treated deployment as a final step rather than a risk to test early. The plan gave Docker and TensorRT 8 days with only 2 days of slack, and I did not try the compiled engine on the target GPU until the pipeline was finished. It finished 26 days late and pushed Station 2 and the handover with it, the main reason the project ended 18 working days late. I also kept no written log, so I noticed slips through conversation instead of measuring them. I learned to test the riskiest item first, while there is still time to respond, and that a simple weekly record costs far less than rebuilding one afterwards.

Signed: _____

Ullas N

I owned the cap and fill model iterations, the first integrated demo, the PatchCore baseline, the EfficientAD label models with calibration, and the GPU verification and handover, 240 hours in total. The label line's journey through three methods showed me how much of the schedule depends on results that cannot be known in advance. Moving to per-region EfficientAD and grayscale input raised crop-level

recall above 92% on every region, though ROI 4 still gives too many false positives, and we report that as a known weakness. Leading the GPU verification, with support from Sushanth and Shreyas, was a useful last check that the stations worked together on the target machine.

One thing I got wrong, and what I learned about project management from it: I planned model work as if each training run would mostly succeed, when iteration was the real task. The cap package finished 15 days late because of small caps, class mapping and colour bias. I also trusted crop-level results for too long: the early 21.4% end-to-end recall exposed stale weights only at test time. I learned to put iteration into the estimate as planned work, to set a rule for when to change method, and to run an end-to-end check early, while problems are still cheap to fix.

Signed: _____

Shreyas Bairy K S

I owned the platform scaffold, the dashboards, the label annotation and localisation package including template matching, and the label dashboard, 220 hours in total. Pushing the scaffold in March gave the team a shared base, so the backend, model and interface work had somewhere to connect. The dashboard work later had the most slack in our plan, which gave me room to support other packages, and I also helped on the fill dataset and the first integrated demo. Working across these packages showed me how much of a project depends on the parts that link other people's work together.

One thing I got wrong, and what I learned about project management from it: I went to a YOLO ROI detector for label localisation before testing a simpler option. Template matching eventually replaced it, and the package finished 12 days late on the critical path, using more hours than planned. I also treated the slack on the label dashboard as spare time: it used 15 of its 20 days. That did not move the finish, but it left little cover for any later slip. I learned to try the simplest workable method first, with a time limit, and to treat slack as a reserve the whole team should know about.

Signed: _____

Consistency Checks

All calculated figures in this report are generated from one data file, so the sections cannot disagree.

Check	Where it comes from	Figure	Agrees?
Work package budgets add up to the project TBC	Table 3.3 = Table 5.1	₹1,74,500	Yes
CBC at the Status Date	Table 5.2 = Table 5.5	₹1,74,500	Yes
CAC includes committed amounts	Table 5.3 = Table 5.5	₹1,89,500 (committed ₹0)	Yes
CEV uses the same percent complete as the status report	Table 5.4 = Section 1	₹1,74,500 (100%)	Yes
Critical path duration equals the planned project duration	Table 4.2 = Figure 4.1	173 days	Yes
Every activity's total slack is consistent with the critical path	Table 4.2	14 activities at 0; none negative	Yes
Contingency reserve appears in the budget	Table 5.1	₹17,450	Yes
Each RAM row has exactly one P	Table 3.4	20 of 20 rows	Yes

Table 10.1 - Consistency checks. The build script fails if any of these does not hold.

Appendix A: WBS Diagram

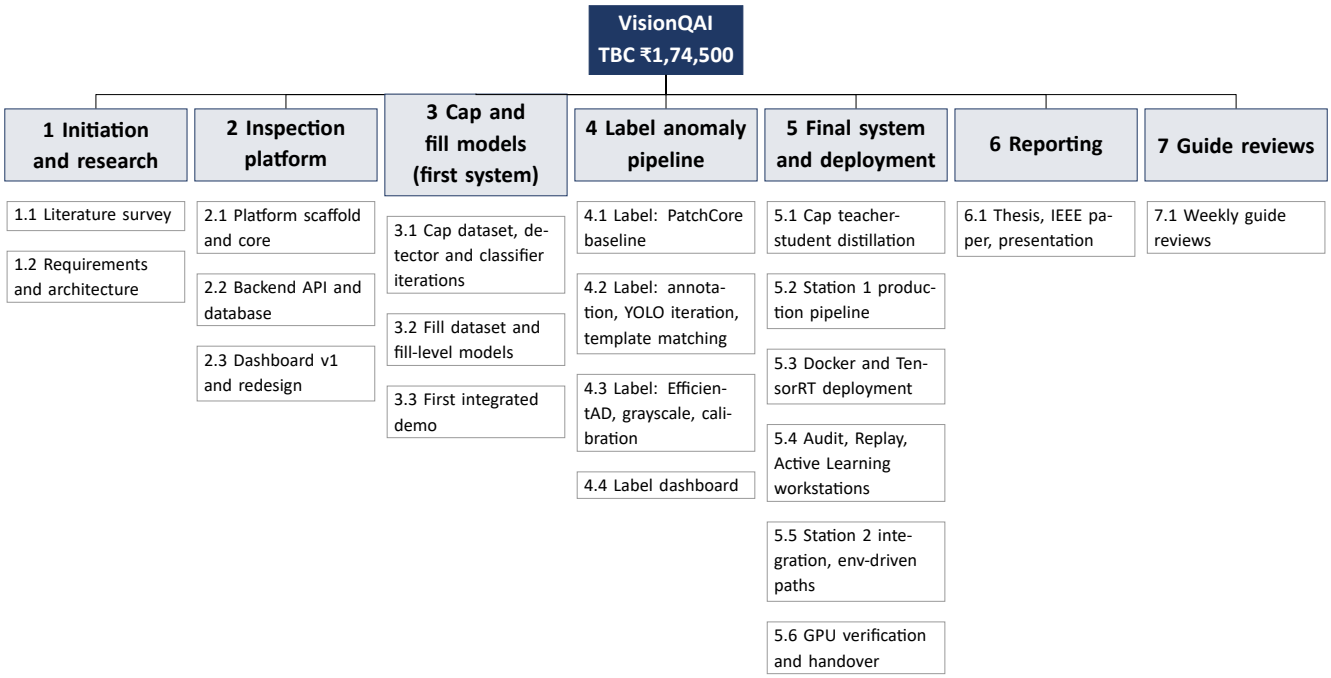


Figure A.1 - Work breakdown structure

Appendix B: Network Diagram and Detailed Gantt Chart

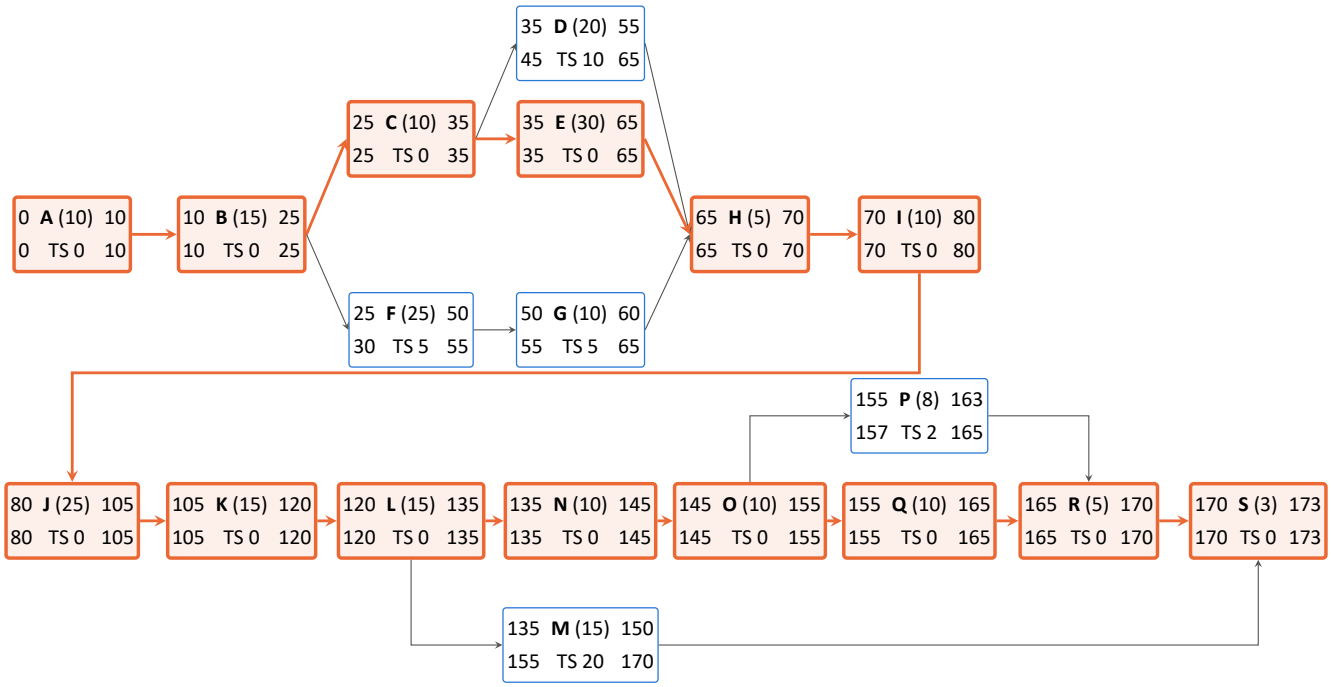


Figure B.1 - Activity-on-node network. Top row of each node: ES, activity (duration), EF. Bottom row: LS, total slack, LF. Critical activities have thick orange borders.

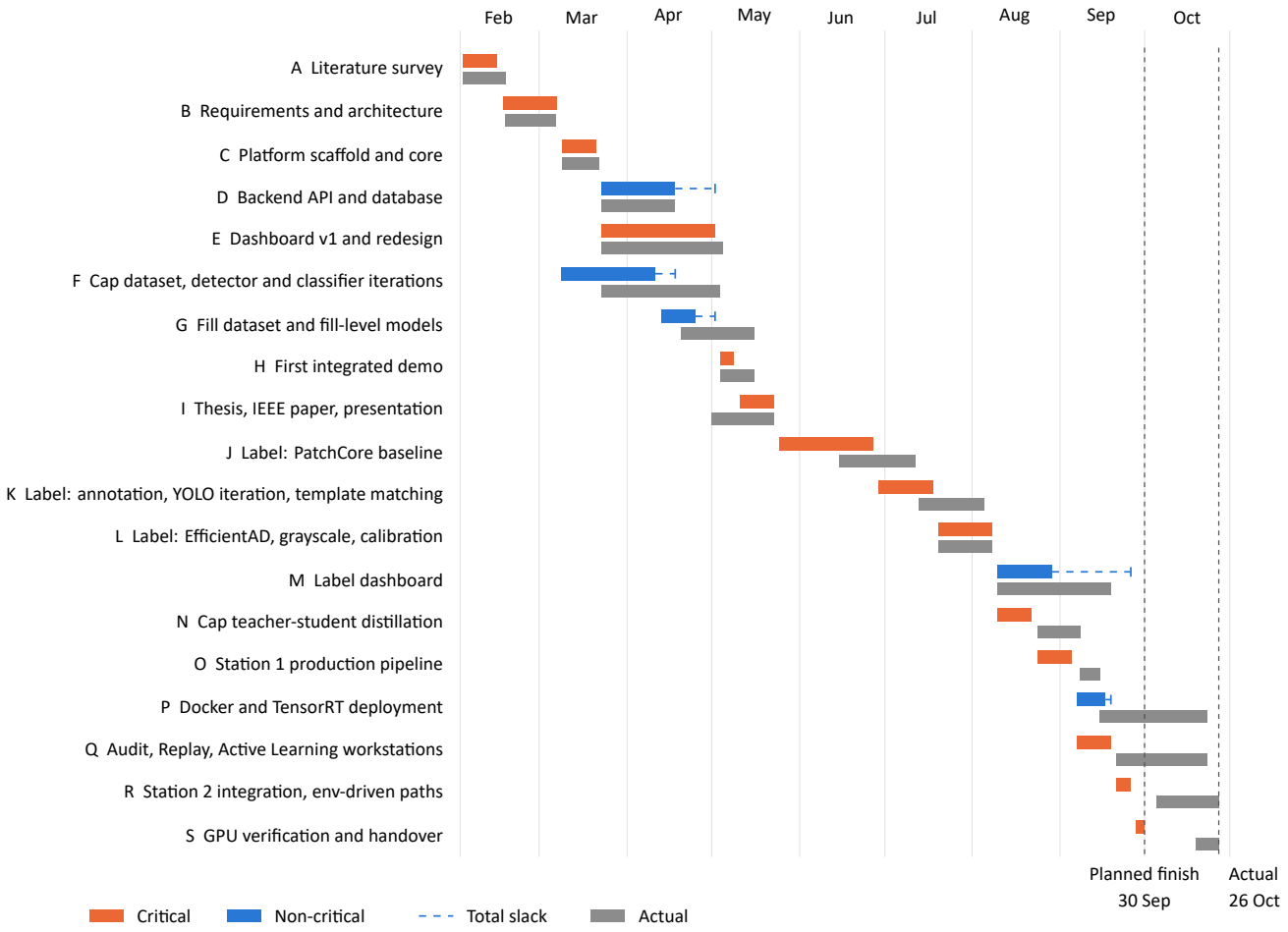


Figure B.2 - Baseline (upper bar) against actual (grey lower bar) for each activity

Appendix C: Status Summaries

These monthly summaries are reconstructed from the Git history for this report; they are not minutes written at the time. Commit counts cover all branches and total 157.

Month	Commits	Status
Feb	0	Literature survey and requirements; nothing pushed yet
Mar	4	Platform scaffold pushed 21 Mar; code review fixes
Apr	9	Execution plan, APIs and database schema, frontend, first cap detection work
May	54	Cap and fill iterations, frontend redesign, integrated demo 15 May, thesis 22 May
Jun	6	Paper update; PatchCore training for the NSC label data
Jul	48	PatchCore concluded; per-region EfficientAD; YOLO ROI and grayscale experiments
Aug	16	Thresholds calibrated, inference fixes; image generator; CSITSS rejection and resubmission plan
Sep	15	Teacher and student training; full pipeline 14 Sep; Docker; TensorRT batch fix; last label commit
Oct	5	Workstations, dynamic batching, stress test, Station 2 integration 26 Oct

Table C.1 - Monthly status from the commit record

Appendix D: Cost Workings, Quotations and Receipts

Cash spending was ₹0, so there are no invoices, quotations or receipts. The GPU and training facility were provided by the college, and datasets came from open-source platforms and Sathwik Solutions.

Notional workings, Status Date: Sushanth 250 h × ₹250; Ullas 240 h × ₹200; Shreyas 220 h × ₹200; guide 35 h × ₹1,000 = ₹35,000. Together, these make the CAC of ₹1,89,500.

Appendix E: Test Results and Other Project Artifacts

Item	Recorded result	Source
Student cap model (validation, final epoch of 126)	Mask mAP50 0.995; mask mAP50-95 0.941; precision 0.996; recall 0.996	results.csv
Cap dataset	8,530 images: train 7,464, valid 712, test 354; 6 classes	TRAINING_PLAN.md
Label, grayscale, crop level	ROI_1 recall 93.9%, precision 83.8%; ROI_2 92.9%, 68.4%; ROI_3 93.5%, 69.0%; ROI_4 92.3%, 30.0% (56 false positives)	PROJECT_CONTEXT.md
Label, end-to-end (early)	Recall 21.4%, accuracy 62.5%; traced to stale YOLOv8 weights in a superseded design	PROJECT_CONTEXT.md
Stress test, CPU fallback only (10 iterations, 19 Oct)	Inference p95 4,443 ms; total cycle p95 4,891 ms; 0 DB write failures; verdict REVIEW. Not a GPU result	stress_test_report.json
GPU verification	Both stations and swap verified by team demonstration on target GPU. No latency figures were retained	Team record
CSITSS 2026 review	Rejected; all three reviewers Borderline; led to the resubmission plan	resubmission_plan.md

Table E.1 - Recorded results